

U.S. Futures and Options Package

For Business: Tuesday, August 27, 2019

Contents:

Cheapest to Deliver Summary Report
Basis Reference Sheet
Treasury Futures Roll Report
Treasury Futures Hot Run Basis Report
Treasury Futures CTD Basis Convergence Report
Treasury Futures Butterfly Report
CBOT Swap Futures
Cross Market Futures Matrix
Cross Market Volatility Spreads Report
Short-Dated Volatility Reports
3Yx10Y Swaption Volatility Model
Volatility Summary Report
Option Market Summary Report
Volatility Skew Model Report
Vega Supply and Demand
Eurodollar Futures Relative Value Report
Asset Swap Spread Cycle
10-Year Swap Spread Model
Swap Spread Curve Report
Super TEDs
Fed Tightening Report
Fed Funds Futures Relative Value
1-Month LIBOR, Treasury Bond and Note Futures
Carry-Efficient Trades Report
Money Market Futures Calendar Spread Report
Eurodollar Butterflies Report
Treasury Yield Curve Report
Maturity Structure of Treasury Volatility
Treasury OTC and Exchange Volatility Report
Treasury Bill TED Analysis Report

Contacts:

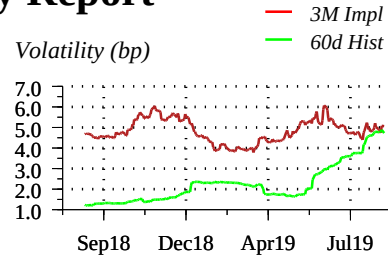
Alex Roever	New York	(212) 834-3316	Joshua Younger	New York	(212) 270-1323
-------------	----------	----------------	----------------	----------	----------------

general email fo.tech@jpmorgan.com

Short-Dated Swaption Volatility Report

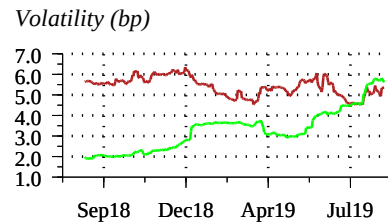
Options on 1yr Swaps

Mat	Implied Vol.		5d	20d	Historical BP Vol.		
	%	bp			10d	20d	60d
1m	50.73	5.47	0.11	1.52	5.16	6.07	4.73
3m	50.75	5.06	0.14	1.13	*1dc	1wc	2wc
6m	53.85	4.87	0.09	0.87	6.07	4.84	3.75
12m	56.95	4.56	-0.06	0.49			



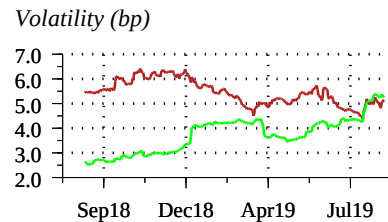
Options on 2yr Swaps

Mat	Implied Vol.		5d	20d	Historical BP Vol.		
	%	bp			10d	20d	60d
1m	62.52	5.88	0.25	1.76	6.21	7.42	5.61
3m	60.03	5.36	0.23	1.28	*1dc	1wc	2wc
6m	60.64	5.07	0.14	0.97	7.42	5.82	4.32
12m	58.77	4.59	-0.07	0.45			



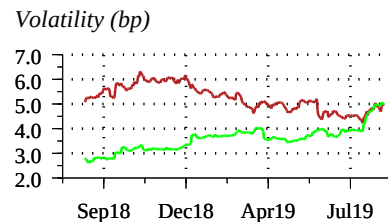
Options on 5yr Swaps

Mat	Implied Vol.		5d	20d	Historical BP Vol.		
	%	bp			10d	20d	60d
1m	66.59	5.71	0.14	1.71	6.06	6.96	5.25
3m	61.32	5.15	0.16	1.14	*1dc	1wc	2wc
6m	58.43	4.80	0.12	0.78	6.96	5.89	4.57
12m	54.91	4.44	0.06	0.38			



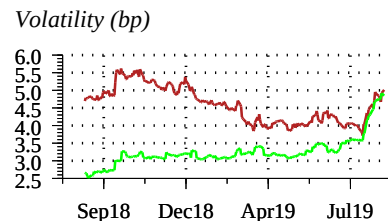
Options on 10yr Swaps

Mat	Implied Vol.		5d	20d	Historical BP Vol.		
	%	bp			10d	20d	60d
1m	60.24	5.52	0.15	1.90	6.14	6.66	4.97
3m	54.98	5.00	0.17	1.32	*1dc	1wc	2wc
6m	51.68	4.68	0.10	0.88	6.66	5.98	4.83
12m	47.98	4.35	0.06	0.49			

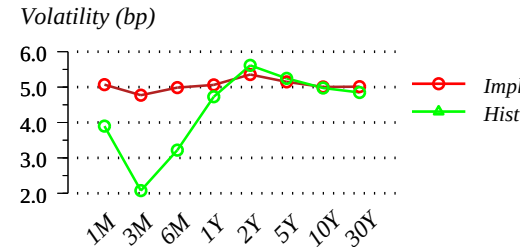


Options on 30yr Swaps

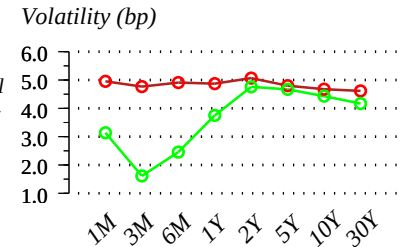
Mat	Implied Vol.		5d	20d	Historical BP Vol.		
	%	bp			10d	20d	60d
1m	53.73	5.59	0.17	2.32	6.61	6.59	4.85
3m	48.37	5.01	0.18	1.65	*1dc	1wc	2wc
6m	44.67	4.61	0.12	1.16	6.59	6.27	5.20
12m	40.96	4.22	0.06	0.73			



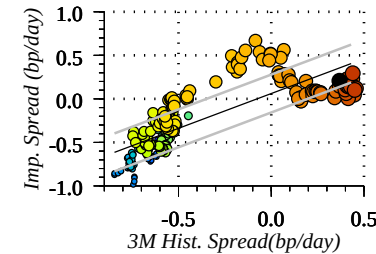
Maturity Structure of 3M Vol.



Maturity Structure of 6M Vol.

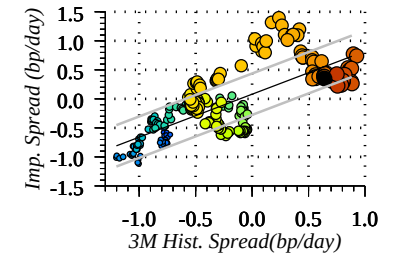


3Mx2Y - 3Mx5Y



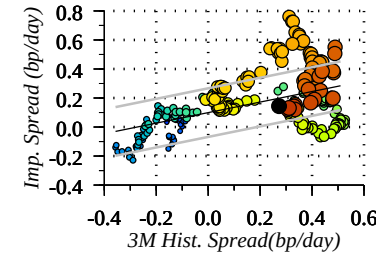
$$Y = 0.7992 X1 + 0.0612 \text{ (adjR2} = 0.6276)$$

3Mx2Y - 3Mx10Y



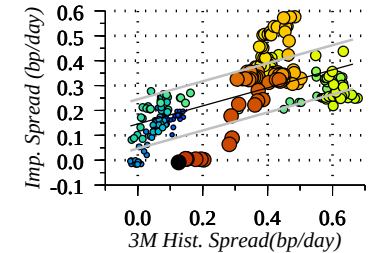
$$Y = 0.7400 X1 + 0.0801 \text{ (adjR2} = 0.5647)$$

3Mx5Y - 3Mx10Y



$$Y = 0.3630 X1 + 0.0993 \text{ (adjR2} = 0.2508)$$

3Mx10y - 3Mx30Y



$$Y = 0.3627 X1 + 0.1458 \text{ (adjR2} = 0.3748)$$

1. Regression of difference in 3M implieds vs difference in 3M historicals.
2. *One-month realized BP Vol. over 1-day, 1-week and 2-week change.

Mat/Und	60d Historical BP Vol.				Implied BP Vol.			
	tight	ease	hold	curr	tight	ease	hold	curr
3mx1y	N/A	N/A	1.56	4.73	N/A	N/A	2.51	5.06
3mx2y	N/A	N/A	2.62	5.61	N/A	N/A	3.34	5.36
3mx5y	N/A	N/A	4.32	5.25	N/A	N/A	4.78	5.15
3mx10y	N/A	N/A	4.76	4.97	N/A	N/A	5.18	5.00
3mx30y	N/A	N/A	4.80	4.85	N/A	N/A	5.08	5.01

Derivatives Strategy

jpmorganmarkets.com

	Eurodollar 1Yr MidCurve					Eurodollar 2Yr MidCurve				Eurodollar 3Yr MidCurve				Eurodollar 4Yr MidCurve				Eurodollar 5Yr MidCurve			
	Sep 19	Oct 19	Nov 19	Dec 19	Mar 20	Sep 19	Oct 19	Nov 19	Dec 19	Sep 19	Oct 19	Nov 19	Dec 19	Sep 19	Oct 19	Nov 19	Dec 19	Sep 19	Oct 19	Nov 19	Dec 19
VOLATILITY																					
Implied (%)																					
Current	73.54	75.53	72.02	70.51	66.23	79.15	75.66	71.28	69.08	77.95	71.90	68.21	67.22	71.63	72.92	68.06	66.19	-	-	-	-
Change (1d)	-11.74	-3.40	-2.10	-2.05	-1.42	-6.70	-2.09	-1.16	-1.25	-1.55	-2.68	-1.50	0.35	-1.99	-1.85	-1.84	-1.74	-	-	-	-
20 day High	86.69	81.30	74.82	73.60	69.80	86.38	79.55	73.47	72.06	84.63	78.90	71.02	70.48	76.57	76.66	70.03	68.92	-	-	-	-
20 day Low	40.47	40.70	61.93	39.44	40.55	40.57	38.92	64.86	37.07	36.28	36.49	59.24	35.69	34.03	34.17	55.36	32.33	-	-	-	-
Business Day	71.74	73.93	71.81	70.47	66.57	77.21	74.06	71.07	69.04	76.04	70.38	68.01	67.18	69.87	71.38	67.86	66.14	-	-	-	-
Historical (%)																					
10 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
50 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2Yr Event Risk	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (%)																					
3 Year High	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Average	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Low	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
% Historic Above Implied	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
BASIS POINT VOLATILITY																					
Implied (bp)																					
Current	6.3	6.1	5.8	5.6	5.1	6.1	5.9	5.6	5.4	6.2	5.8	5.5	5.4	6.1	6.0	5.6	5.4	-	-	-	-
Change (1d)	-0.6	-0.2	-0.1	-0.1	-0.1	-0.5	-0.1	-0.1	-0.1	-0.1	-0.2	-0.1	0.1	-0.1	-0.1	-0.1	-0.1	-	-	-	-
Change (5d)	-0.2	0.2	0.2	0.3	0.1	-0.2	0.1	0.1	0.1	0.1	0.0	0.0	0.2	-0.2	0.3	0.1	0.2	-	-	-	-
Change (20d)	1.4	1.3	-	1.0	0.8	1.6	1.4	-	1.0	1.9	1.4	-	1.3	1.8	1.8	-	1.3	-	-	-	-
20 day High	6.9	6.4	5.9	5.8	5.3	6.7	6.2	5.7	5.6	6.7	6.3	5.7	5.6	6.5	6.2	5.8	5.6	-	-	-	-
20 day Low	4.4	4.5	5.4	4.3	4.2	4.2	4.2	5.2	4.0	4.0	4.0	5.1	3.9	4.0	4.1	5.1	4.0	-	-	-	-
Historical (bp)																					
10 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
50 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (bp)																					
3 Year High	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Average	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Low	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
% Historic Above Implied	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Yield Change Required to... (bp)																					
Offset 1 C-day Decay	4.9	5.1	4.8	4.7	4.1	4.8	4.8	4.5	4.4	5.1	4.8	4.5	4.5	4.8	5.1	4.7	4.6	-	-	-	-
Offset 1 B-day Decay	5.9	6.1	5.8	5.7	5.0	5.8	5.7	5.4	5.2	6.2	5.8	5.5	5.4	5.8	6.2	5.7	5.6	-	-	-	-
Offset 1 Month Decay	-	24.9	24.4	24.0	20.9	-	22.1	22.1	21.7	-	23.4	22.9	22.7	-	26.2	24.6	23.9	-	-	-	-
Break Even at Expiry	18.0	26.8	33.9	38.2	47.6	17.6	26.6	33.1	37.1	17.4	25.7	32.1	36.7	17.4	26.2	32.6	36.7	-	-	-	-
BPV per Contract	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00
Cal/Bus Days to Expiry	18/13	46/33	81/56	109/75	200/136	18/13	46/33	81/56	109/75	18/13	46/33	81/56	109/75	18/13	46/33	81/56	109/75	18/13	46/33	81/56	109/75
Futures Price	98.675	98.695	98.695	98.695	98.775	98.790	98.765	98.765	98.765	98.725	98.705	98.705	98.705	98.670	98.645	98.645	98.645	98.590	98.560	98.560	98.560

VOLATILITY is price volatility for bond, note and index options; yield volatility for money market options.

Volatility Cones are based on number of days to expiration. For example, if 40 days remain, we look at actual volatility of the nearby contract in every 40 day period over the last 3 years.

% Historic Above Implied shows how often actual volatility exceeded current implied volatility. A higher number means cheaper options.

BASIS POINT VOLATILITY is basis point yield volatility for all options.

Yield Change Required to... is based on an at-the-money straddle. Yield change for bond, note and money market options; Price change for equity and commodity options.